

Level 14: Over-The-Air (OTA) Live Updates & Runtime Versioning

React Native Masterclass

Expo SDK 54 / RN 0.81

Master Certification

14.1 — What is an OTA Update?

React Native mobile apps consist of two layers: the **Native Binary Shell** (C++/Java/Swift) and the **JavaScript Bundle** (`index.bundle`). While native binary modifications require 2-3 days of Apple/Google App Store reviews, JavaScript logic, UI components, and styling can be updated **Over-The-Air in seconds** via `eas update` .

```
import * as Updates from 'expo-updates';

const checkAndApplyLiveUpdates = async () => {
  const update = await Updates.checkForUpdateAsync();
  if (update.isAvailable) {
    await Updates.fetchUpdateAsync();
    await Updates.reloadAsync(); // Instant live update!
  }
};
```

🧐 Tenglish Explanation:

App Store approval kosam 3 days aagalsina pani lekunda, JavaScript bug fixes and UI updates ni `eas update` dwara 5 seconds lo users phone loki live ga push cheyochu.

14.2 — Apple & Google Store Legal Policies

Update Type	Status	Policy Rule
Bug Fixes & UI Updates	100% LEGAL ✅	Explicitly allowed under Apple Guideline 2.5.2 & Google Play policies.
Core Purpose Shift (e.g. Casino)	ILLEGAL ❌	Results in immediate permanent account termination.
Bypassing In-App Purchases	ILLEGAL ❌	Bypassing Apple/Google digital payment fees results in store ban.
New Native Modules (C++/Swift)	REQUIRES BINARY ⚠️	OTA cannot deliver native code; requires full store binary upload.

14.3 — Runtime Versioning & Native Compatibility Locks

To prevent JavaScript code from invoking native C++/Java methods that don't exist in older app binaries, `runtimeVersion` creates a cryptographic compatibility lock:

```
// app.json
{
  "expo": {
    "runtimeVersion": { "policy": "appVersion" },
    "updates": { "url": "https://u.expo.dev/your-project-id" }
  }
}
```

Tenglish Explanation:

User phone lo patha native code unnapudu kotha JS code velli app crash aypokunda, `runtimeVersion` lock chesi exact matching version unna phones ki mathrame update download cheyisthundi.

14.4 — Emergency Rollbacks, Channels & Canary Releases

Deployment Channels: `eas update --channel preview` for QA testing; `eas update --channel production` for live App Store users.

Emergency 1-Second Rollback: `eas update:republish --group-id <stable-group-id>` instantly reverts all production devices to the previous working build.

Tenglish Summary:

Production lo sudden bug unte `eas update:republish` dwara 1 second lo patha stable version ki rollback (undo) chestham. Idi enterprise mobile architecture  complete mastery!